

Causality: Basics (Part I)

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Roadmap

The lecture is organized into three parts:

- Part I** Basic concepts of causality and an overview of causal frameworks
- Part II** Introduction to two main frameworks (PO and SCM) with simple examples of causal effect identification and estimation
- Part III** Overview of various causal inference methods and their key characteristics

Part I introduces the foundations of causal reasoning:

- What causality is and why it matters
- How causation differs fundamentally from statistical association
- The need for formal frameworks and assumptions in causal inference

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What is Causality?

Definition

"Causality is **influence** by which **one** event, state, or object (a cause) **contributes to the production of another** (an effect)

where the cause is **partly** responsible for the effect, and the effect is **partly** dependent on the cause."¹

(cause) $\implies$ (effect)

- : 'increases', 'decreases' vs. linked to, associated with
- token causality (specific event) vs. type causality (general property)

¹ modified from Wikipedia <https://en.wikipedia.org/wiki/Causality>

Why Do We Study Causality?

- Definition of **Science** 
 - ▶ "Knowledge or a system of knowledge covering general truths or **the operation of general laws** especially as obtained and tested through scientific method." 
- Laws = Causal mechanism
- Causality in various academic disciplines 
 - ▶ Natural Sciences: Physics, Chemistry , Biology, Climate Science ,
 - ▶ Psychology , Social Science , Economics ,
 - ▶ Epidemiology , Public Health 
- How about Engineering  where precise control matters.

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How is Causality related to & {ML, AI, & DS}?

Machine Learning

Mainly focused on learning correlation, e.g., $\hat{P}_{\theta}(y | \mathbf{x}) \approx P(y | \mathbf{x})$
But what does it ... learn? *association is not causation.*

Artificial Intelligence

a rational agent performing actions to achieve a goal
e.g., Reinforcement Learning $\pi_{\theta}(\text{action} | \text{state})$

Data Science

Capture, Process, Analyze (e.g., Stat, ML), Communicate with Data
How can causal conclusions be drawn from data?

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Pearl's Causal Hierarchy

- Level 1: 👁 Associational or Observational
- Level 2: 🛠 Interventional or Experimental
- Level 3: 🤖 Counterfactual (🛠 | 👁)

The term 'counterfactual' is used a bit differently depending on the context.

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Correlation (Level 1) vs. Causation (Level 2): Simpson's Paradox

Consider the following scenario:

1. A Patient with **Kidney Stone** 🏥 visits a Hospital 🏥.
2. A Doctor examines the Patient and provides a **Treatment** 🏥.
3. The Patient's Health Outcome 📊 is later reported 📊.

 **Dataset w/ 3 variables:**

stone $\in \{\text{small, large}\}$, **treatment** $\in \{A, B\}$, **outcome** $\in \{T, F\}$

Simpson's Paradox

		Treatment	
		A	B
Stone	Small	Group 1	Group 2
		93% (81/ 87)	87% (234/270)
	Large	Group 3	Group 4
		73% (192/263)	69% (55/ 80)

Each cell represents $P(\text{success} \mid \text{treatment}, \text{stone})$

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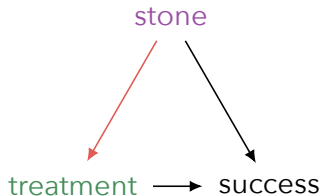
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Simpson's Paradox

		Treatment	
		A	B
Stone	p 2		
		4/270)	
	p 4		
		5/ 80)	
		9/350)	

Stone



what-we-have = observational ☹

Each cell represents

Aggregated: $P(\text{succ})$

Simpson's Paradox

		Treatment	
		A	B
Stone	p 2		
		4/270)	
	p 4		
		5/ 80)	
Aggregated:		9/350)	

Stone

do
treatment → success

stone



what-we-want = $\mathcal{P}_{\star}$ interventional

Each cell represents

Aggregated: $P(\text{succ})$

Simpson's Paradox

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		A	B
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Treatment is *selected* based on individual traits.

What if we administer each treatment *randomly* 🎲?

Simpson's Paradox

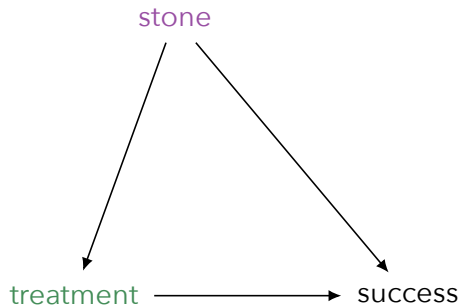
		Treatment	
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Stone	Small	Group 1 93% (81/ 87)	Group 2 87% (234/270)
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Efficacy		83.2%	78.2%

What if we administer each treatment *randomly* ? **The causal effect of A**

$$P(\text{succ} \mid \text{do}(A)) = \sum_{\text{stone}} P(\text{succ} \mid A, \text{stone})P(\text{stone}) = 0.832$$

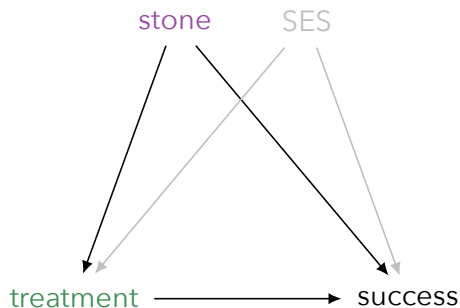
Caveat: Unmeasured Confounding

Confounding Variables



Caveat: Unmeasured Confounding

Unobserved Confounding Variables



Lessons Learned from Simpson's Paradox

- Causal analyses need to be guided by subject-matter knowledge .
- Identical data arising from different causal structures need to be analysed differently.
- No purely statistical rules exist to guide causal analyses.

Data & Questions

Anyone dealing with data should take care of the types of **data** and **question**:

		Question 	
		Associational	Causal
 Data	Observational	Observational Study, Machine Learning*	Causal Inference
	Experimental	Causal Inference	Experimental, A/B testing, RL

*ML, in general, does not care about the type of data but the question should match with the data type.

Formalizing Causality: Desiderata

The goal is to develop a **model** or **framework** that is sufficiently expressive to **formalize** fundamental intuitions about causal relationships.

1. Need a $\frac{\neq}{\equiv}^+$ mathematical framework (=model) of unknown reality. 🌐
2. It should support the ③ levels of causal quantities.
3. There should be a way to impose (causal) constraints about the reality.

Formalizing Causality

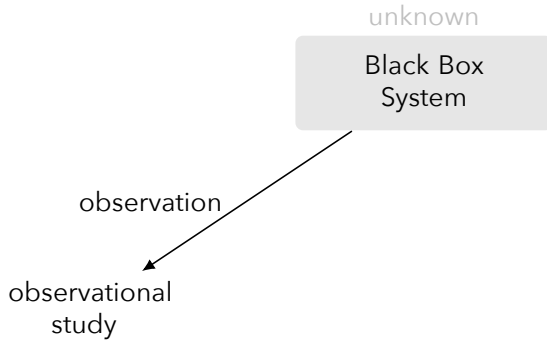
Observation & Intervention (Experiments)

unknown

Black Box
System

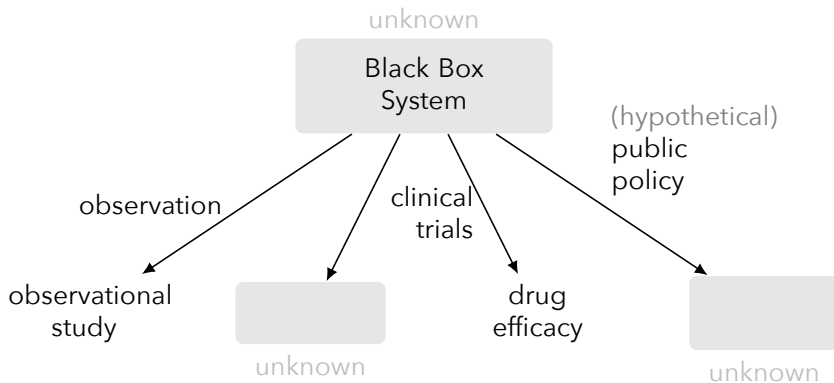
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Formalizing Causality

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Frameworks for Causal Inference

- Two commonly used frameworks
 - ▶ **Potential Outcome framework** (PO), also known as the counterfactual framework, or the Neyman-Rubin Causal Model (Neyman, 1923; Rubin, 1974),
 - ▶ **Structural Causal Model** framework (Pearl, 2000)
- Mathematically the two frameworks are *connected*.

Potential Outcome Framework

- The **Potential Outcomes (PO) Framework** is one of the most *widely adopted* approaches to causal inference, especially in fields such as medicine, public health, economics, and the social sciences.
- The framework traces its origin to *Neyman* (1923) for randomized experiments and was later extended by *Rubin* (1974) to observational studies.
- It formulates causal inference as a missing data problem, where only one potential outcome is observed per unit.

Potential Outcomes

- **Potential Outcomes***: The values of a **unit**'s measurement of interest after
 - (a) application of the **treatment** and
 - (b) non-application of the treatment (i.e., under control)
- e.g., Patient's Success after A? or under B?
 - ▶ **Unit**: An individual entity (e.g., person, place, or object) observed at a specific time. Note that the same entity at different times is considered as distinct units.
 - ▶ **Treatment**: An intervention applied to units, whose effect on a particular outcome is assessed relative to a control or alternative condition.

Potential Outcomes and Missing Data

- $Y(0)$ denotes the outcome given the **control** treatment, and $Y(1)$ the outcome given the **active treatment**
- The fundamental problem of causal inference: We can observe at most one of the potential outcomes for each unit, the other(s) are missing.

Unit	Potential Outcomes		Observed	
	$Y(x=0)$	$Y(x=1)$	X	Y
1	0	1	1	
2	0	1	0	
3	0	0	1	
4	1	1	1	
5	1	0	0	

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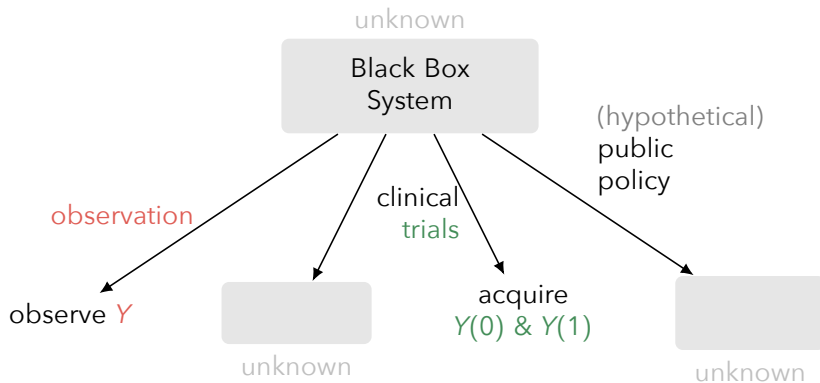
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Unit	Potential Outcomes		Observed	
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1	?	1	1	1
2	0	?	0	0
3	?	0	1	0
4	?	1	1	1
5	1	?	0	1

Formalizing Causality: Potential Outcome



We later make **assumptions** on the **system** using **potential outcomes**.

Structural Causal Model

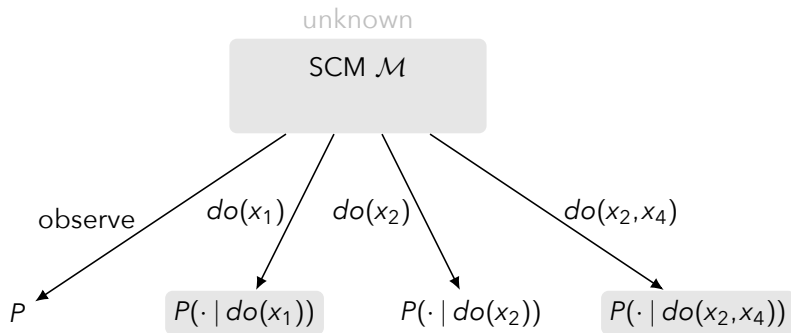
- **Represents causal relationships structurally:** each **variable** as a deterministic function of its **direct causes** (parents) and **exogenous noises**:

$$X_i = f_i(\text{PA}_i, U_i)$$

- **Encodes causal assumptions via graphs:** The model induces a *Directed Acyclic Graph* that visually captures the causal structure.
- **Explicitly models interventions:** Using the *do-operator* $do(X = x)$, SCMs define interventions by modifying structural equations.
- **PO focuses on estimation, not model structure:** The PO framework *traditionally* emphasizes estimands and statistical estimation rather than modeling the data-generating process explicitly.

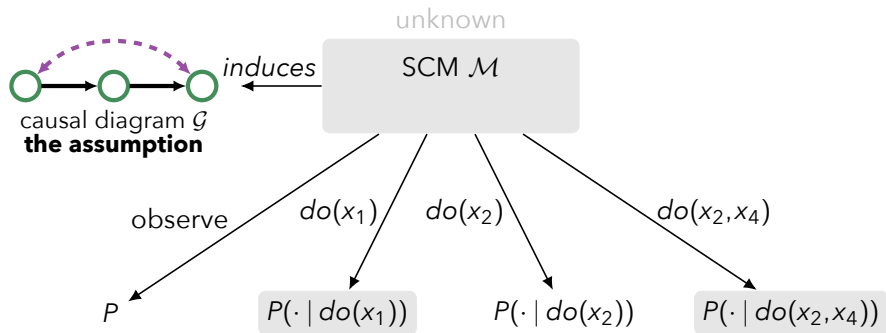
Formalizing Causality: Structural Causal Model

SCM for Black Box, Intervention as $do(\cdot)$



Formalizing Causality: Structural Causal Model

SCM for Black Box, Intervention as $do(\cdot)$, and Graph for assumptions



Structural Causal Models

mathematical understanding of **unknown reality**

Definition (Structural Causal Model)

An SCM $\mathcal{M}$ is a 4-tuple $\langle \mathbf{V}, \mathbf{U}, \mathbf{F}, P(\mathbf{U}) \rangle$

- $\mathbf{V} = \{V_1, \dots, V_n\}$ are **endogenous** variables;
- $\mathbf{U}$ is a set of **exogenous** variables;
- $\mathbf{V}$ are determined by functions $\mathbf{F} = \{f_1, \dots, f_n\}$,

$$v_i \leftarrow f_i(\mathbf{pa}_i, \mathbf{u}_i)$$

where $\mathbf{pa}_i \subseteq \mathbf{V} \setminus \{V_i\}$, $\mathbf{u}_i \subseteq \mathbf{U}$;

$\mathbf{U}$ can be viewed as what each unit **actually** is about. $\mathbf{U}$ determines $\mathbf{V}$

Structural Causal Models

mathematical understanding of **unknown reality**

Definition (Structural Causal Model)

An SCM $\mathcal{M}$ is $\langle \mathbf{V}, \mathbf{U}, \mathbf{F}, P(\mathbf{U}) \rangle$

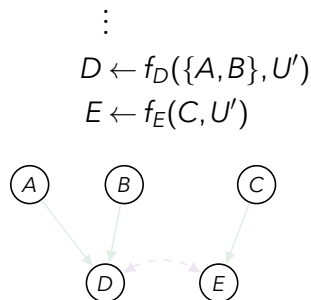
- $\mathbf{V}$ are **measured** and $\mathbf{U}$ are **unknown**;
- While **unknowns** $\mathbf{U} \sim P(\mathbf{U})$
Each **known variable** is determined by an (unknown) function,

$$v \leftarrow f(\text{some known}, \text{some unknown})$$

$\mathbf{U}$ can be viewed as what each unit **actually** is about. $\mathbf{U}$ determines $\mathbf{V}$

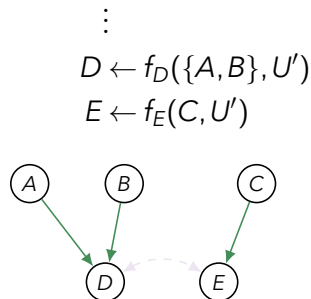
Model $\Rightarrow$ Causal Diagram

- SCM induces a **causal diagram** $\mathcal{G}$ based on the **signature of function**,
- Represented as a Directed Acyclic Graph:
 1. (Nodes) Each $V \in \mathbf{V}$ is a node,
 2. (**Directed** Edge: Causal Relationship) $B \rightarrow D$ since B is the argument of the function of D .
 3. (**Bidirected** Edge: Unobserved Confounders) $D \longleftrightarrow E$ if $\mathbf{U}_D$ and $\mathbf{U}_E$ correlate (here, $\mathbf{U}_D = \mathbf{U}_E = \{U'\}$).



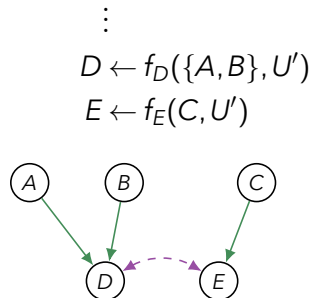
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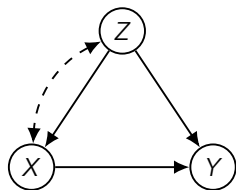
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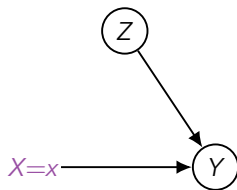
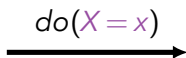


Intervention – $do(\cdot)$ operator

- fixing variable $X \in \mathbf{V}$ to a constant value x is denoted using as $do(X = x)$.
- These two graphs represent **model** *before* and *after* $\mathcal{I}_X do(X = x)$.



Causal Graph $\mathcal{G}$

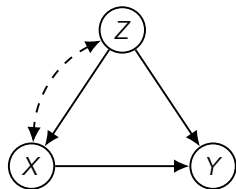


Causal Graph under Intervention $\mathcal{G}_{\overline{X}}$

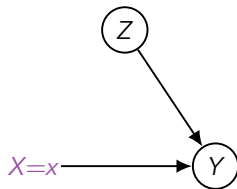
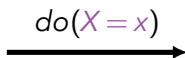
- $do(\cdot)$ operator induces a submodel $\mathcal{M}_x$ with $f_x \in \mathbf{F}$ replaced to a constant x .

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Causal Graph under Intervention $\mathcal{G}_{\bar{X}}$

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Summary

- **Causality** is essential for understanding and guiding decisions in complex systems.
- **Causal frameworks and assumptions** are critical for valid inference.
- **Potential Outcomes (PO)** define causal effects as comparisons between outcomes under different treatments.
- **Structural Causal Models (SCM)** represent the data-generating process structurally, supporting interventions and counterfactual reasoning.